Autonomous DevOps

Setup Guide

Complete step-by-step guide to install, configure, and run the Self-Healing CI/CD Agent

Version: 1.0.0

Date: July 28, 2026

License: MIT

1. Prerequisites

Before you begin, ensure you have the following installed on your system:

> Python 3.10+ -- Download from python.org or use pyenv
> Node.js 18+ -- Download from nodejs.org or use nvm
> Docker 24+ -- Download from docker.com (required for sandbox verification)
> Git -- Usually pre-installed; apt install git / brew install git

API Keys Required

Service	Purpose	Where to Get
OpenAI / OpenRouter	LLM for code analysis & fixes	platform.openai.com/api-keys
GitHub PAT	Repo access & PR creation	github.com/settings/tokens
Langfuse (optional)	Observability & tracing	langfuse.com

2. Quick Start

2.1 Clone the Repository

```
git clone https://github.com/Sarancoding/Autonomous-DevOps.git
cd Autonomous-DevOps
```

2.2 Backend Setup

```
cd backend
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

2.3 Frontend Setup

```
cd frontend
npm install          # or: bun install
npm run dev          # or: bun dev
```

2.4 Configure Environment

```
# Create .env file:
LLM_API_KEY=sk-your-key-here
GITHUB_TOKEN=ghp_your-token-here
GITHUB_WEBHOOK_SECRET=your-webhook-secret
```

2.5 Run with Docker Compose

```
docker compose up --build
```

This starts: Backend (localhost:8000), Frontend (localhost:5173), Redis.

3. GitHub Webhook Configuration

For automatic triggering on CI failures:

> Go to Settings -> Webhooks -> Add webhook
> Payload URL: https://your-domain.com/webhook/github
> Content type: application/json
> Secret: Match GITHUB_WEBHOOK_SECRET
> Events: Check runs, Push events, Workflow runs

[AI] The agent will automatically parse failures, generate fixes, verify in Docker, and submit a PR.

4. Dashboard Walkthrough

Open http://localhost:5173 and explore:

> Settings: Enter LLM API Key and GitHub Token (BYOK)
> Dashboard: Paste repo URL + failure log, click 'Run Agent'
> Job Detail: View timeline, diff, live WebSocket logs
> Analytics: Track token usage, costs, success rates

5. Troubleshooting

Problem	Solution
Backend won't start	Set LLM_API_KEY environment variable
Docker sandbox fails	Ensure Docker is running: docker info
WebSocket not connecting	Check CORS origins in config
GitHub token issue	Ensure PAT has 'repo' scope
LLM API key invalid	Verify key has GPT-4o access

6. Security Best Practices

- BYOK: API keys sent via headers, never persisted to disk
- Zero retention: Source code deleted after job completes
- Sandbox isolation: Each job in isolated Docker container, no network
- HMAC verification: Webhook signatures validated on every request
- Audit logs: All agent actions logged with timestamps

7. License

This project is licensed under the MIT License.